

Geometry

Unit 6

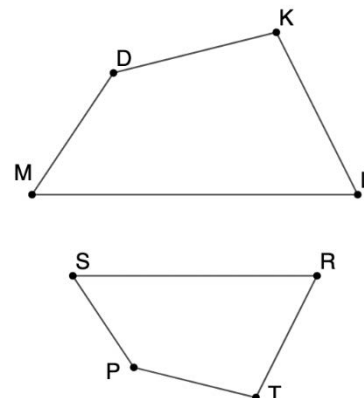
Lesson 11 Homework

Name _____

Date _____

Period _____

$DKLM$ and $PTRS$ are shown, where $DKLM \sim PTRS$ with $MD = 20$, $DK = 6z + 6$, $KL = 8w + 1$, $LM = 40$, $PT = 12$, $TR = 4w - 2$, and $RS = 16$.



- Determine the value of each variable.

$$w = \underline{\hspace{2cm}}$$

$$z = \underline{\hspace{2cm}}$$

- Determine the value of each side.

$$DK = \underline{\hspace{2cm}}$$

$$SP = \underline{\hspace{2cm}}$$

$$TR = \underline{\hspace{2cm}}$$

- What is the perimeter of $DKLM$?

Consider additional measurements of angles for $DKLM$ and $PTRS$ where $m\angle DKL = 150^\circ$, $m\angle MDK = (15x - 25)^\circ$, $m\angle SPT = (12x + 5)^\circ$, $m\angle DML = (5y - 5)^\circ$, $m\angle PSR = (4y + 4)^\circ$.

- Determine the value of each variable.

$$x = \underline{\hspace{2cm}}$$

$$y = \underline{\hspace{2cm}}$$

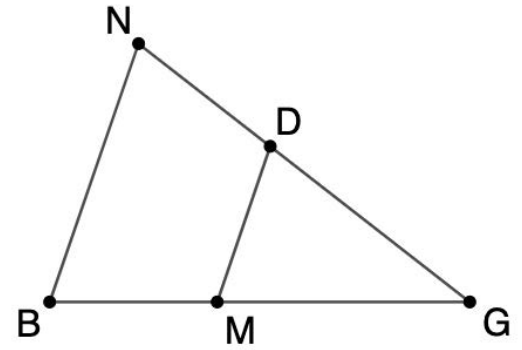
- Determine the value of each angle.

$$m\angle MDK = \underline{\hspace{2cm}}$$

$$m\angle PTR = \underline{\hspace{2cm}}$$

$$m\angle PSR = \underline{\hspace{2cm}}$$

$\triangle NBG$ and $\triangle DMG$ are shown, where $\triangle NBG \sim \triangle DMG$ with $DM = 8$, $NB = 32$, $DG = 7$, $MG = 2y - 8$, and $BG = 4y + 4$.



6. What is the scale factor from $\triangle DMG$ to $\triangle NBG$?

7. What is the value of the variable, y ?

8. What is the length of BG ?

Consider additional measurements of angles for $\triangle NBG$ and $\triangle DMG$ where $m\angle GNB = (8z + 3)^\circ$, $m\angle DMG = (7w - 9)^\circ$, $m\angle NBG = (6w + 3)^\circ$, $m\angle GDM = (10z - 15)^\circ$.

9. Determine the value of each variable.

$w = \underline{\hspace{2cm}}$

$z = \underline{\hspace{2cm}}$

10. Determine the value of each angle.

$m\angle GNB = \underline{\hspace{2cm}}$

$m\angle GBN = \underline{\hspace{2cm}}$

$m\angle DGM = \underline{\hspace{2cm}}$